

Umeshkumar Ghaskata

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University at Buffalo, SUNY - MS Computer Science

New York, USA

Indian Institute of Technology(IIT), Bombay - B.Tech Engineering

Mumbai, India

SCHOLASTIC ACHIEVEMENT

- Ranked in the **top 0.7% of 1.3+ Million candidates** in nation wide compatatave entrance examination [2020]
- Achieved **97.75 percentile** in JEE Advanced among 250,000 selected candidates [2020]

EXPERIENCE

Roswell Park Comprehensive Cancer Center — Software Engineer

NY, United States [Jun'2025-May'2026]

- Fine-tuned an open-source **LLM (Gemma3-1B)** on an **AWS Cloud Linux** instance using **PyTorch, Transformers and Quantized LoRA**. Architected a personalized AI coaching chatbot using a (RAG), grounding the model's responses.
- Hosted fine-tuned LLM using **vLLM** to Private VPC, desgined Low latency API to eliminate reliance on **OpenAI API**
- Build a custom **MCP** server acting as a adapter, securely connecting AI assistant directly to users' private NoSQL databases.
- Owned the end-to-end integration of the **AI backend** with a cross-platform React Native frontend, enabling real-time, personalized conversational coaching for users.
- Managed source code and collaborative development using **Git**, implemented **CI/CD pipelines** with GitHub Actions
- Implemented **Redis semantic caching** and rate limiting in a **Node.js Express backend**, reducing redundant API requests.
- Followed **Agile practices** through weekly meetings with user, advisor : taking **direction and constructive criticism**.

DRONES Lab (NASA AIST-21 Project) — Research Assistant

NY, United States [Aug'2024-Mar'2026]

- Trained, evaluated, and deployed **Deep Learning models** (PCA+MLP, FPGA-based architectures) for hyperspectral image classification improving land-cover recognition accuracy from 75.4% to 94.5%.
- Reduced **edge inference latency** from **295ms to 52ms** using optimized Model to enable real-time classification.
- Engineered benchmarking DL model across edge devices **RTX 4060, Jetson Orin, and Xavier** guided Hardware selection.
- Published Thesis** on energy-efficient coverage path planning; developed a **Python-based** optimization framework.

Scitara Corporation — Software Engineer

Mumbai, India [Nov'2022-Feb'2023]

- Developed low-latency **RESTful APIs** for **SaaS Product** using **Docker and Kubernetes (AWS EKS)**, establishing a highly available infrastructure capable of supporting **high-throughput** AI inference workloads that reduced testing time.
- Facilitate **backend** with **FastAPI Microservices** with **JWT Authentication** and MockAPI service with **PostgreSQL**, equipped with **Kubernetes(AWS EKS)** for production.

PROJECTS

Multimodal AI Platform with RAG, Streaming

GitHub  [Jan'2026–Mar'2026]

OpenAI/Claude API, pgvector, FastAPI, ragas, DeepEval, Docker

- Architected a multimodal RAG platform using **FastAPI** and **Supabase (pgvector)**, implementing advanced semantic **chunking strategies** and **dense embeddings** to optimize document indexing and retrieval.
- Engineered an **automated evaluation** pipeline integrating **ragas** and **DeepEval** to benchmark **context precision**.
- Designed a LLM retrieval pipeline utilizing **OpenAI/Claude APIs** and multimodal semantic search, optimizing vector queries to reduce response time while streaming answers to the frontend.

Enterprise Multi-Agent Orchestration System

[Apr'2026–Present]

LangGraph, FastAPI, RabbitMQ, WebSockets, PostgreSQL, Llama 3.3, LangSmith, DeepEval

- Architected an asynchronous multi-agent system using **LangGraph** and **FastAPI**, orchestrating specialized worker agents to scrape live web data and synthesize complex plans using **Llama 3.3**, with **PostgreSQL** handling state persistence.
- Engineered a decoupled backend architecture utilizing **RabbitMQ** background task queues to process long-running agent workflows, while streaming intermediate agent reasoning to the frontend via **WebSockets**.

TECHNICAL SKILLS SUMMARY

Programming: Python, TypeScript, JavaScript, SQL, C++

AI Engineering & Agents: LangGraph, LangChain, Model Context Protocol (MCP), RAG, vLLM, Tool Calling

LLMs, Eval & Observability: Llama 3.3, Claude 3.5, GPT-4o, DeepEval, ragas, LangSmith, Prometheus, Grafana

Backend & Message Queues: FastAPI, Node.js, WebSockets, RabbitMQ, Kafka, Celery, gRPC, REST API

Databases & Vector Stores: PostgreSQL, pgvector, Redis, MongoDB, Supabase, Pinecone

Cloud, DevOps & Infra: AWS, GCP, Docker, Kubernetes (EKS), Terraform, CI/CD, GitHub Actions

Frontend & Mobile: React.js, React Native, HTML/CSS, Tailwind

Certification: Google Generative AI Leader [\[Badge Link\]](#)